

# **Customer, Product, and Profitability Performance Analysis in Supply Chain Operations**

*An Exploratory Data Analysis and Profitability Intelligence Approach for Supply Chain Decision-Making*

## **Research Paper**

Submitted by

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## ABSTRACT

Supply chain organizations generate substantial transactional data through customer orders, products, pricing, discounts, and geographic markets, yet high sales volume does not necessarily translate into high profitability. This study presents an exploratory profitability analysis of an APL Logistics (KWE Group) order-item dataset comprising 180,519 records and 40 variables, covering 20,652 unique customers, 118 products, 50 product categories, five markets, and 23 order regions. Following data-quality validation and financial-consistency checks, the analysis evaluates profitability across customer, product, category, discount, and geographic dimensions using indicators including profit margin, customer value index, category margin, loss rate, and a defined discount impact ratio.

The dataset records total sales of 36.78 million and total profit of 3.97 million, an overall profit margin of 10.78%. Profitability is highly concentrated at the product level, with the top 10 products contributing approximately 89.70% of total profit, while the top 10 customers contribute only about 0.50%. Approximately 19.70% of customers are individually loss-making despite generating a meaningful share of total sales. Discount analysis shows profit margin declining from 13.09% on non-discounted orders to 9.39% within the 20–25% discount band, while market analysis shows Europe leading in commercial scale and USCA recording the highest market-level margin at 11.14%.

The study concludes that profitability decisions should draw on customer, product, discount, and geographic performance jointly rather than on revenue or order volume alone. An interactive Streamlit dashboard was developed to operationalize these findings for ongoing decision support.

**Keywords:** Customer Profitability; Discount Impact; Profitability Analysis; Supply Chain Analytics; Supply Chain Operations

## 1. INTRODUCTION

### 1.1 Background

Modern supply chain organizations generate extensive transactional data through customer orders, products, sales, pricing, discounts, and geographic markets, creating an opportunity to move beyond basic reporting toward data-driven decision-making. A persistent challenge, however, is that revenue and profitability do not always move together: a customer can generate high sales while contributing relatively low profit, and a product may show strong demand while operating at a comparatively weak margin. Discounting and regional differences add further variation to profitability outcomes.

Customer profitability research reinforces this concern. Reinartz and Kumar (2000) found that long-lived customers are not necessarily the most profitable customers, demonstrating that financial contribution, rather than relationship duration or transaction volume, should guide customer evaluation. For supply chain organizations specifically, profitability analysis therefore requires a multidimensional perspective in which customer contribution, product economics, discount exposure, and market performance are considered together rather than in isolation.

This project adopts such a perspective by analyzing transactional order-item data associated with APL Logistics (KWE Group), using exploratory data analysis to identify areas of financial strength, margin pressure, and potential business review.

### 1.2 Business Problem

The central business concern motivating this study is that high revenue does not automatically represent high business value. Organizational leadership requires visibility into which customers, products, categories, and markets contribute meaningful profit and which areas carry hidden profitability risk — including high-revenue but low-margin products, loss-making customers, discount levels associated with margin compression, and regional profitability differences. The objective is not simply to identify the largest sources of revenue, but to determine where revenue is accompanied by meaningful profit.

### 1.3 Research Objectives

The objectives of this study are to:

1. Validate and understand the available order, customer, product, sales, profit, discount, and geographic data.
2. Calculate total revenue, total profit, and overall profit margin as a financial baseline.
3. Evaluate customer-level contribution and identify loss-making customers using quartile-based value tiers.
4. Analyze product- and category-level profitability, including high-revenue, low-margin products.
5. Examine the relationship between discount levels and profit margin.
6. Compare profitability across markets, regions, and customer countries.
7. Assess whether delivery and shipping performance are associated with order-level profitability.
8. Translate the findings into actionable business recommendations.
9. Implement an interactive Streamlit dashboard for ongoing decision support.

### 1.4 Significance of the Study

This study provides a data-driven framework for identifying the true drivers of profitability rather than relying on revenue as a proxy for performance. By evaluating customer, product, discount, and market dimensions together and operationalizing the findings through an interactive dashboard, it offers both an analytical contribution and a practical decision-support tool for pricing, discount governance, and customer and portfolio strategy.

## 2. LITERATURE REVIEW

### 2.1 Customer Profitability Analysis

Customer profitability analysis emphasizes the financial contribution of individual customers rather than sales volume or relationship duration. Traditional customer management can overemphasize revenue or transaction frequency, yet these do not necessarily represent profitability. Reinartz and Kumar (2000) demonstrated that long-term customers are not automatically the most profitable, underscoring the importance of evaluating customer economics directly. For supply chain organizations, this matters because customers differ in order frequency, product mix, discount exposure, and transaction-level profitability — a customer generating substantial sales may contribute less profit than a smaller but more consistently profitable one, motivating the customer value-tier analysis undertaken in this study.

### 2.2 Product and Category Profitability

Product-level analysis complements customer profitability, since revenue concentration does not imply equal profitability across products: a portfolio may contain a small number of items responsible for most total profit, while other high-volume products generate weaker margins. Jahani, Jain, and Ivanov (2023) note that analytics methods applied to supply chain

and logistics data increasingly support this kind of granular, item-level performance diagnosis rather than aggregate reporting alone. This motivates the product- and category-level analysis, and the identification of high-revenue, low-margin products, undertaken in this study.

### 2.3 Discounting and Margin Management

Discounting is commonly used to stimulate demand or support competitive positioning, but it can also reduce revenue retained per transaction. Kumar, Madan, and Srinivasan (2004) note that promotion profitability depends on customer response, sales expansion, and pricing-mechanism structure, so discount decisions should not be evaluated solely by their effect on sales volume. This study compares margin and loss rate across discount bands to identify associations warranting review, without asserting causality.

### 2.4 Data-Driven Decision-Making in Supply Chains

Data-driven decision-making enables organizations to integrate transactional information with analytical methods to support more informed managerial decisions. Brynjolfsson, Hitt, and Kim (2011) reported a positive relationship between data-driven decision-making and firm performance. Within supply chain management specifically, big-data analytics can support improved visibility, forecasting, and performance management: Gopal et al. (2022) examined factors influencing this impact, while Jahani, Jain, and Ivanov (2023) reviewed analytics methodologies applied in supply chain and logistics research — together supporting the rationale for combining financial, customer, product, and geographic information within a single analytical framework.

### 2.5 Research Gap

Existing research provides strong foundations across customer profitability, promotional decision-making, and supply chain analytics, while this study brings them together within a single exploratory framework. This study combines customer, product, discount, and geographic performance in one analysis, extended through an interactive Streamlit dashboard that moves the findings from static reporting toward practical decision support.

## 3. DATASET AND METHODOLOGY

### 3.1 Dataset Overview

The dataset contains transactional order-item-level information related to customers, products, sales, profitability, discounts, geography, and shipping operations, sourced from the APL Logistics (KWE Group) project made available through the internship. Table 1 summarizes the scale and key dimensions of the dataset.

**Table 1. Dataset Overview**

| Dataset Attribute                                                       | Value   |
|-------------------------------------------------------------------------|---------|
| Order-item records                                                      | 180,519 |
| Variables                                                               | 40      |
| Unique customers                                                        | 20,652  |
| Unique products                                                         | 118     |
| Product categories                                                      | 50      |
| Markets                                                                 | 5       |
| Order regions                                                           | 23      |
| Customer countries meeting analysis threshold ( $\geq 100$ order items) | 2       |

The five markets represented in the dataset are Europe, LATAM, Pacific Asia, USCA, and Africa. Customer segments are Consumer, Corporate, and Home Office, and shipping information includes Standard Class, Second Class, First Class, and Same Day modes.

### 3.2 Key Analytical Variables

The 40 variables were grouped conceptually rather than examined individually: financial variables (Sales, Order Item Total, Order Profit Per Order, Order Item Discount, Discount Rate) support revenue and profitability measurement; product variables (Product Name, Category Name, Product Price) support product- and category-level analysis; customer variables (Customer Id, Customer Segment, Customer Country) support customer aggregation; geographic variables (Market, Order Region) support market-level comparison; and logistics variables (Delivery Status, Days for Shipping, Shipping Mode) support the operational analysis.

### 3.3 Data Cleaning and Validation

Before performing profitability analysis, the dataset was audited for missing values, duplicate records, financial consistency, and problematic zero or negative values. The audit identified 11 missing values in total, limited to two fields — 8 in Customer Lname and 3 in Customer Zipcode — with no duplicate records. These missing values were confined to descriptive customer fields rather than the financial variables used in profitability calculations, and were therefore not material to the analysis.

Financial consistency was checked using the relationship  $\text{Order Item Total} \approx \text{Sales} - \text{Order Item Discount}$ , applying a 0.01 rounding tolerance: 175,906 records showed an exact match and 4,613 showed a one-cent difference, with none exceeding tolerance. These one-cent differences were treated as source-data rounding effects rather than data-quality errors, and no correction was applied.

No negative values were found in Sales, Order Item Total, product prices, quantity, discount, or shipping-duration fields. Negative values in Order Profit Per Order were retained because they represent legitimate loss-making transactions, and excluding them would have understated the true extent of unprofitable activity. The Benefit per order field was found identical to Order Profit Per Order across all 180,519 records and was treated as redundant. The dataset lacks a dedicated currency field, so monetary values are reported in the dataset's original units.

### 3.4 Analytical Workflow

The analysis followed a structured sequence beginning with the raw dataset and proceeding through data audit and cleaning, financial validation, KPI calculation, customer/product/discount/market analysis, and business interpretation, culminating in the Streamlit dashboard. Figure 1 illustrates this workflow, which ensures that business conclusions are derived only after data quality and financial consistency have been established.

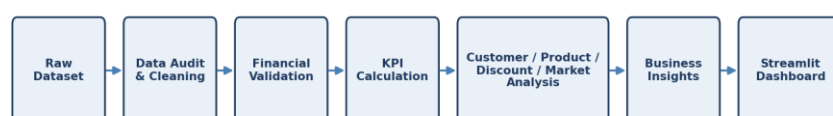


Figure 1. Analytical Workflow

### 3.5 Key Performance Indicators

The following indicators were defined and used consistently throughout the analysis.

#### **Total Revenue and Total Profit**

$$\text{Total Revenue} = \Sigma \text{Sales} \quad \text{Total Profit} = \Sigma \text{Order Profit Per Order}$$

#### **Profit Margin**

$$\text{Profit Margin (\%)} = (\text{Total Profit} \div \text{Total Sales}) \times 100$$

#### **Customer Value Index**

Defined for this study as the average total profit contribution per customer:

$$\text{Customer Value Index} = \text{Total Profit} \div \text{Number of Unique Customers}$$

The resulting Customer Value Index is approximately 192.08 monetary units per customer. This is a project-specific analytical definition rather than a standardized industry metric.

#### **Category Margin and Loss Rate**

Each individual category's margin is calculated as:

$$\text{Category Margin} = (\text{Category Profit} \div \text{Category Sales}) \times 100$$

For the overall KPI, this study reports the mean of the 50 individual category-level margins, rather than a single sales-weighted margin across all categories:

$$\text{Category Margin KPI (\%)} = \text{Mean of Category-Level Margins, across all 50 categories}$$

This averaging produces a Category Margin KPI of approximately 10.76%, close to but distinct from the overall business margin of 10.78% described in Section 3.5. Because this KPI averages category margins without weighting by category sales volume, it should not be expected to exactly equal the aggregate business margin, and is best read as an indicator of typical category-level profitability rather than a restatement of total profit divided by total sales.

$$\text{Loss Rate (\%)} = (\text{Loss-Making Order Items} \div \text{Total Order Items}) \times 100$$

#### **Discount Impact Ratio**

The Discount Impact Ratio measures the relative reduction in profit margin between non-discounted and discounted order items:

$$\text{Discount Impact Ratio (\%)} = ((\text{No-Discount Margin} - \text{Discounted Margin}) \div \text{No-Discount Margin}) \times 100$$

Using the observed margins of 13.09% for non-discounted order items and 10.65% for discounted order items, the overall Discount Impact Ratio is approximately 18.67%. This should be interpreted as a relative reduction in margin associated with discounted orders under this study's defined calculation, and not as a statement that 18.67% of total profit was lost.

### **3.6 Analytical Approach and Tools**

The analysis applies descriptive and exploratory methods — aggregation by customer, product, category, market, and region; profit-margin and loss-rate calculation; quartile-based segmentation; discount-band and high-revenue/low-margin identification; and geographic and shipping-performance comparison. All relationships are interpreted as associations rather than causal effects. Data preparation and exploratory analysis were implemented in Python using pandas and NumPy for data wrangling (McKinney, 2022) and Matplotlib and Seaborn for visualization, within Jupyter Notebook, with Streamlit used to operationalize the findings through an interactive dashboard.

## **4. RESULTS AND ANALYSIS**

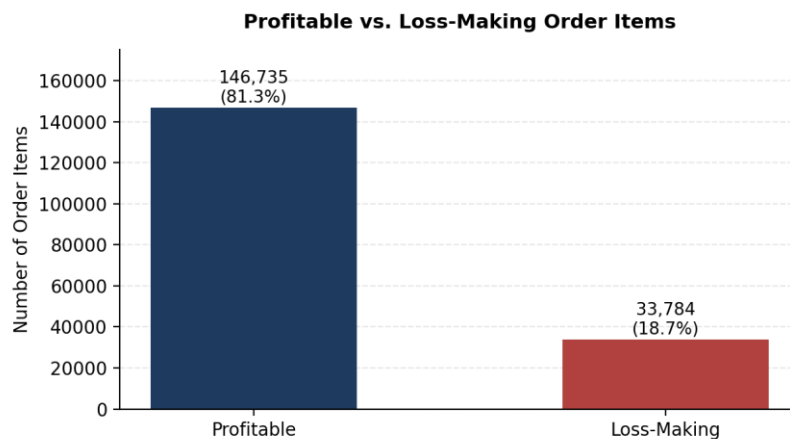
### **4.1 Revenue and Profitability Overview**

The dataset generates total sales of 36,784,734.31 and total profit of 3,966,902.97, resulting in an overall profit margin of 10.78%. This provides the financial baseline for all subsequent analysis, indicating that approximately 10.78% of dataset-reported sales value is represented by profit according to the available profit field. Table 2 summarizes the overall business KPIs.

**Table 2. Overall Business KPIs**

| KPI                   | Result        |
|-----------------------|---------------|
| Total Sales           | 36,784,734.31 |
| Total Profit          | 3,966,902.97  |
| Overall Profit Margin | 10.78%        |
| Unique Customers      | 20,652        |
| Unique Products       | 118           |
| Product Categories    | 50            |
| Customer Value Index  | 192.08        |

Of the 180,519 order items, 146,735 ( $\approx 81.3\%$ ) are individually profitable and 33,784 ( $\approx 18.7\%$ ) are loss-making, as shown in Figure 2. This aggregate margin conceals substantial variation across customers, products, categories, discounts, and markets, so overall revenue and profit are useful starting points but insufficient alone for identifying the underlying sources of profitability. Because the dataset lacks a usable order-date field, comparisons across discount bands and other groups are presented cross-sectionally rather than as time trends.



**Figure 2. Profitable vs. Loss-Making Order Items**

## 4.2 Customer Contribution and Profitability

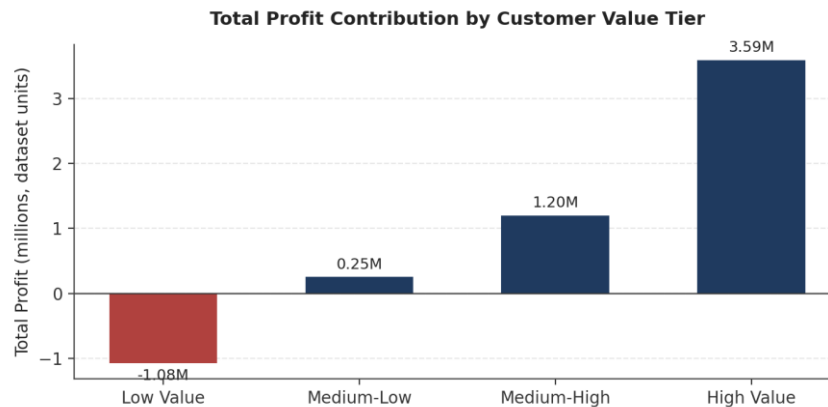
Customer-level analysis, performed by aggregating sales and profit by Customer Id, identifies 4,069 loss-making customers out of 20,652 ( $\approx 19.70\%$  of the customer base). These customers collectively generated approximately 7,114,079.34 in sales (19.34% of total sales) while contributing approximately  $-1,080,299.72$  in total profit — demonstrating why customer evaluation based on revenue alone can be misleading.

Customers were further divided into four quartile-based value tiers using total customer profit. Table 3 and Figure 3 summarize the resulting segmentation.

**Table 3. Customer Value Tiers**

| Customer Tier | Customers | Total Sales  | Total Profit    | Avg. Profit |
|---------------|-----------|--------------|-----------------|-------------|
| Low Value     | 5,163     | 7,236,946.40 | $-1,077,122.61$ | $-208.62$   |
| Medium-Low    | 5,165     | 2,895,952.27 | 251,273.72      | 48.65       |

| Customer Tier | Customers | Total Sales   | Total Profit | Avg. Profit |
|---------------|-----------|---------------|--------------|-------------|
| Medium-High   | 5,161     | 8,949,535.70  | 1,201,134.77 | 232.73      |
| High Value    | 5,163     | 17,702,299.94 | 3,591,617.09 | 695.65      |

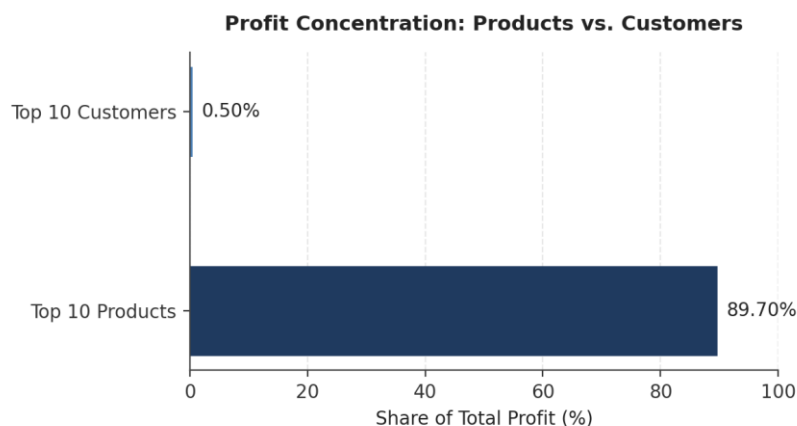


**Figure 3. Total Profit Contribution by Customer Value Tier**

These tiers are statistical quartiles based on customer profit rather than predefined business segments, identifying relative profitability levels within the population. The High Value tier generates the largest total sales and profit contribution, while the Low Value tier produces negative aggregate profit. Notably, the top 10 customers contribute only approximately 0.50% of total profit (Section 4.3) — customer profitability is therefore comparatively dispersed, in clear contrast to the concentration observed at the product level.

#### 4.3 Product and Category Profitability

Product-level analysis reveals strong concentration in profitability: the top 10 products generate approximately 3,558,207.42 in profit, equivalent to approximately 89.70% of total business profit. Figure 4 contrasts this concentration with the comparatively dispersed profitability of the top 10 customers (0.50% of total profit), illustrating that product-level and customer-level profitability require different management approaches.



**Figure 4. Profit Concentration: Products vs. Customers**

Category-level analysis further demonstrates substantial variation in profit margin. Among categories with at least 100 order items, the strongest margins include Fitness Accessories (14.77%), Soccer (14.74%), Women's Clothing (13.62%), Baseball & Softball (13.57%), and Garden (12.97%), while the weakest include Strength Training (0.61%), Men's

Clothing (4.57%), Books (7.02%), Video Games (8.16%), and DVDs (8.38%). High transaction volume should therefore not automatically be read as strong category economics.

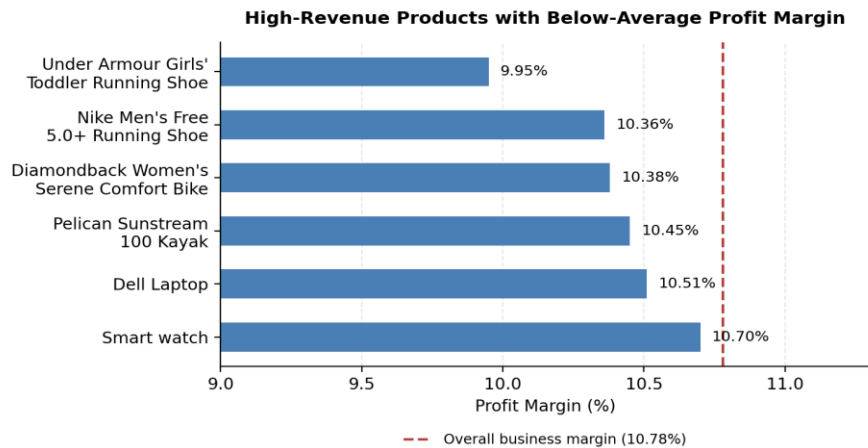
Loss exposure at the transaction level also varies considerably. Every category with at least 100 order items remains net profitable in aggregate, yet the highest individual loss rates occur in Men's Clothing (24.04%), Health and Beauty (23.48%), Video Games (21.60%), Golf Shoes (21.18%), and Pet Supplies (21.14%). Category performance should therefore be assessed using total profit, margin, and loss rate together rather than any single measure.

#### **High-Revenue, Low-Margin Products**

A dedicated analysis identified products combining substantial revenue with comparatively weak profitability. Products with at least 100 order items were retained, and a product was classified as high-revenue, low-margin when its sales fell within the upper quartile of the filtered product population and its profit margin was below the overall business margin of 10.78%. Six products met these conditions, summarized in Table 4 and Figure 5.

**Table 4. High-Revenue, Low-Margin Products**

| Product                                              | Total Sales  | Profit Margin | Loss Rate |
|------------------------------------------------------|--------------|---------------|-----------|
| Under Armour Girls' Toddler Spine Surge Running Shoe | 1,269,082.65 | 9.95%         | 19.64%    |
| Nike Men's Free 5.0+ Running Shoe                    | 3,667,633.20 | 10.36%        | 18.66%    |
| Diamondback Women's Serene Classic Comfort Bike      | 4,118,425.42 | 10.38%        | 18.87%    |
| Pelican Sunstream 100 Kayak                          | 3,099,845.00 | 10.45%        | 18.81%    |
| Dell Laptop                                          | 663,000.00   | 10.51%        | 16.52%    |
| Smart watch                                          | 117,006.75   | 10.70%        | 16.81%    |



**Figure 5. High-Revenue Products with Below-Average Profit Margin**

The Nike Men's Free 5.0+ Running Shoe, Diamondback Women's Serene Classic Comfort Bike, and Pelican Sunstream 100 Kayak are particularly notable for combining multi-million-level sales with sub-average margins, while the Under Armour Girls' Toddler Spine Surge Running Shoe records the lowest margin among the six at approximately 9.95%. The analysis does not establish the specific cause; pricing, discounting, product mix, and cost structure should be investigated before corrective action is taken.

#### **4.4 Discount Impact Analysis**

Discounted order items account for 170,491 of 180,519 records ( $\approx 94.4\%$ ). Discounted transactions record an average profit of 21.70 and a margin of 10.65%, versus 26.67 and 13.09% for non-discounted transactions; loss rates are comparatively similar (18.73% vs. 18.38%). The margin difference is therefore more pronounced than the loss-rate difference,

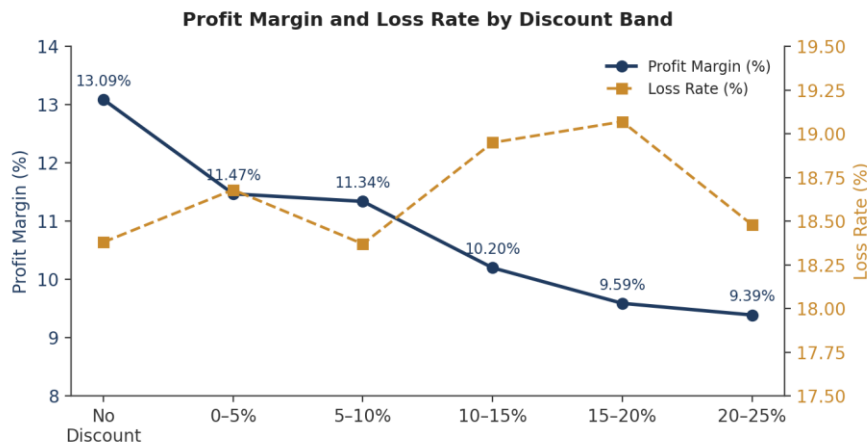
indicating that discounting is associated with lower realized margins while its link to loss frequency is less straightforward.

Order items were further divided into discount bands, summarized in Table 5 and Figure 6. The highest observed discount band is 20–25%; no transactions in the dataset carry a discount above 25%, so no conclusion can be drawn beyond that level.

The analysis also evaluated the average Order Item Profit Ratio — a per-order-item measure already present in the dataset — across the same discount bands, providing an additional profitability lens alongside aggregate margin and loss rate. Table 5 reports both measures.

**Table 5. Profit Margin, Avg. Profit Ratio, and Loss Rate by Discount Band**

| Discount Band | Order Items | Avg. Profit | Avg. Profit Ratio | Profit Margin | Loss Rate |
|---------------|-------------|-------------|-------------------|---------------|-----------|
| No Discount   | 10,028      | 26.67       | 12.75%            | 13.09%        | 18.38%    |
| 0–5%          | 50,143      | 23.37       | 12.10%            | 11.47%        | 18.68%    |
| 5–10%         | 40,116      | 23.10       | 12.27%            | 11.34%        | 18.37%    |
| 10–15%        | 30,087      | 20.78       | 11.79%            | 10.20%        | 18.95%    |
| 15–20%        | 40,116      | 19.54       | 11.69%            | 9.59%         | 19.07%    |
| 20–25%        | 10,029      | 19.13       | 12.72%            | 9.39%         | 18.48%    |



**Figure 6. Profit Margin and Loss Rate by Discount Band**

Profit margin declines steadily from 13.09% with no discount to 9.39% at the 20–25% band, a reduction of approximately 3.71 percentage points. Loss rates, however, do not increase monotonically: the highest loss rate occurs in the 15–20% band (19.07%), while the 20–25% band records a lower 18.48%. The defensible conclusion is therefore that higher discounts are consistently associated with progressive margin compression, while loss rates fluctuate rather than rise with discount depth.

The average Order Item Profit Ratio follows a visibly different pattern from profit margin: it stays comparatively stable, moving narrowly between 11.69% and 12.75%, and rises again to 12.72% in the 20–25% band rather than continuing to fall. This divergence is expected rather than contradictory — Profit Margin is a sales-weighted aggregate (total profit ÷ total sales), so it is pulled down by a small number of high-value, low-margin discounted orders, whereas the Avg. Profit Ratio is an unweighted mean of each order item's individual profit ratio, giving equal weight to every order regardless of size. Together, the two measures indicate that while the largest discounted orders show clear margin compression, the typical individual order item does not show the same degree of profitability erosion at higher discount levels.

The overall Discount Impact Ratio, as defined in Section 3.5, is approximately 18.67%, indicating that the observed margin for discounted transactions is proportionally lower than that of non-discounted transactions; this should be read as a relative margin-reduction indicator rather than a share of total profit lost.

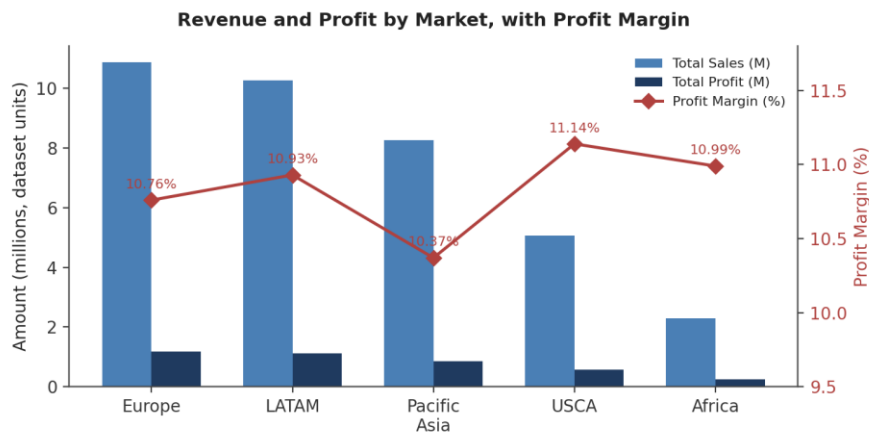
The 20–25% band was further examined at category level (categories with at least 100 order items in that band, using a discount rate between 0.20 and 0.25 inclusive). The weakest margins include Accessories (7.85%), Men's Footwear (8.12%), Shop By Sport (8.16%), Water Sports (8.34%), and Golf Gloves (8.38%); Accessories also records the highest loss rate, at approximately 23.58%. Several high-volume categories — Men's Footwear, Women's Apparel, Indoor/Outdoor Games, Fishing, Cardio Equipment, and Camping & Hiking — each exceed 1,400 high-discount order items while maintaining margins below 9.50%. This suggests discount policy may need to be differentiated by category, though it does not by itself prove discounting alone caused the weaker margins.

#### 4.5 Market and Regional Profitability

Table 6 and Figure 7 summarize profitability at the market level. Europe generates the highest total sales and total profit, reflecting its commercial scale, while USCA records the highest market-level profit margin at 11.14% despite a smaller sales scale than Europe and LATAM. Pacific Asia records the lowest market-level margin (10.37%) and the highest loss rate (18.92%) among the five markets. The differences in loss rate across markets are comparatively narrow, ranging from 18.58% to 18.92%, so the more meaningful market-level differences relate to commercial scale and margin efficiency rather than loss frequency.

**Table 6. Market-Level Profitability**

| Market       | Order Items | Total Sales   | Total Profit | Margin | Loss Rate |
|--------------|-------------|---------------|--------------|--------|-----------|
| Europe       | 50,252      | 10,872,396.60 | 1,169,442.96 | 10.76% | 18.75%    |
| LATAM        | 51,594      | 10,277,612.64 | 1,123,321.61 | 10.93% | 18.58%    |
| Pacific Asia | 41,260      | 8,273,743.58  | 857,753.44   | 10.37% | 18.92%    |
| USCA         | 25,799      | 5,066,528.61  | 564,313.78   | 11.14% | 18.65%    |
| Africa       | 11,614      | 2,294,452.88  | 252,071.18   | 10.99% | 18.58%    |



**Figure 7. Revenue, Profit, and Margin by Market**

At the regional level (23 order regions), Southern Africa records the highest observed profit margin at approximately 26.64%, despite modest sales of approximately 228,251.59; other high-margin regions include Canada (24.92%), Central Asia (23.59%), and East Africa (23.31%). By contrast, Western Europe and Central America generate the largest absolute profit contributions ( $\approx 625,446$  and  $\approx 616,342$  respectively) because of their much larger sales

scale, even though their margins are more moderate — indicating that regional decisions should weigh both absolute contribution and margin efficiency rather than either alone.

Only two customer countries meet the minimum threshold of 100 order items used for country-level analysis: EE. UU. and Puerto Rico. EE. UU. contributes approximately 22.63 million in sales and 2.45 million in profit (margin  $\approx 10.84\%$ ), while Puerto Rico contributes approximately 14.15 million in sales and 1.51 million in profit (margin  $\approx 10.70\%$ ). With only two countries meeting the threshold, these results indicate geographic concentration rather than a comprehensive global comparison.

#### **4.6 Delivery and Shipping Performance**

Delivery status shows only a modest observed link to profitability: Late Delivery accounts for approximately 54.8% of order items, followed by Advance Shipping ( $\approx 23.0\%$ ), Shipping on Time ( $\approx 17.8\%$ ), and Shipping Canceled ( $\approx 4.3\%$ ), while average profit ( $\approx 20.70$ – $22.71$ ) and loss rate ( $\approx 18.54$ – $19.20\%$ ) vary only slightly across these groups.

Shipping performance, derived from the difference between actual and scheduled shipping duration and classified as Early, On Time, or Late, shows a similarly narrow spread: loss rates are 18.67% (Early), 18.57% (On Time), and 18.78% (Late), with comparable average profit ( $\approx 21.59$ – $22.53$ ). Shipping mode shows somewhat larger variation — First Class records an average profit of 23.12 and a loss rate of 18.34%, versus 20.85 and 18.85% for Same Day. Because the dataset lacks a dedicated shipping-cost field, this analysis evaluates the association between shipping performance and profitability rather than its direct financial cost; the small observed differences do not support shipping delay as a major profitability differentiator within this dataset.

### **5. DISCUSSION AND LIMITATIONS**

#### **5.1 Discussion**

The findings provide a multidimensional view of profitability and demonstrate why revenue alone is insufficient for evaluating supply chain performance. The overall business margin of 10.78% offers a useful baseline, but substantial variation exists beneath this figure across every dimension examined.

At the customer level, approximately 19.70% of customers are individually loss-making, yet they still account for approximately 19.34% of total sales — a clear illustration that revenue-generating customers can nonetheless produce negative aggregate profit. At the product level the pattern is markedly different: the top 10 products contribute approximately 89.70% of total profit. Customer and product management therefore require different approaches, since customer profitability is comparatively dispersed while product profitability is highly concentrated.

Category analysis shows that net profitability does not eliminate transaction-level risk, since several categories remain profitable overall while containing loss rates above 20%. Discount analysis provides one of the clearest patterns in the study — margin declines steadily from 13.09% without discounts to 9.39% at 20–25% discount — yet loss rates do not increase monotonically, which prevents the analysis from overstating the relationship between discounting and losses. Geographic analysis shows scale and efficiency to be distinct dimensions: Europe leads in total sales and profit, while USCA achieves the highest market-level margin, and Southern Africa and Canada exhibit strong regional margins despite smaller sales scale.

Operational analysis offers a more moderate result, with shipping delay and delivery status showing comparatively small profitability differences. Within this dataset, discount

governance, customer profitability, and product/category performance therefore appear to be stronger areas for review than shipping delay alone, and the findings support an integrated approach in which management considers all of these dimensions simultaneously.

## 5.2 Limitations

- **Observational data:** the dataset reflects historical transactions rather than a controlled experiment, so relationships between discounts, shipping performance, and profitability should not be read as causal.
- **Missing cost context:** there is no dedicated shipping-cost field or full logistics-cost breakdown, so profitability relies on the available profit field rather than a complete cost-accounting model.
- **Absence of a time dimension:** no usable order-date field exists, so true time-series trends cannot be established; findings are cross-sectional comparisons rather than temporal trends.
- **Small analytical groups:** some products, categories, and discount–category combinations contain few transactions, so minimum thresholds were applied to avoid overinterpretation.
- **Geographic concentration:** only two customer countries meet the minimum threshold for country-level analysis, so those findings indicate concentration rather than global comparison.
- **Currency information:** no dedicated currency field exists, so monetary values are reported in the dataset's original units.
- **Generalizability:** findings are specific to this dataset's variables, population, and period, and should not be generalized without further validation.

## 6. BUSINESS RECOMMENDATIONS

- **Strengthen discount governance:** because profit margin declines consistently across discount bands, discount policies should incorporate profitability thresholds — informed by customer value, product margin, category economics, and expected sales response — rather than a uniform discount approach, with the 20–25% band deserving particular attention as the lowest-margin band observed.
- **Monitor customer profitability:** the 4,069 identified loss-making customers should be reviewed for pricing structure, discount exposure, product mix, and service economics before further strategic investment; customer value overall should be evaluated using actual profit contribution rather than sales alone.
- **Protect high-contribution products:** because the top 10 products contribute approximately 89.70% of total profit, these products warrant close monitoring, since changes in their pricing, discount exposure, availability, or cost structure may have a disproportionately large effect on total business profit.
- **Investigate high-revenue, low-margin products:** the six identified products should be reviewed individually — not necessarily to reduce sales, but to assess whether margin improvement is achievable through pricing adjustments, discount optimization, product-cost review, or improved commercial terms.
- **Apply category-specific discount controls:** categories such as Accessories, Men's Footwear, and Shop By Sport show weak margins under 20–25% discounting and may benefit from tighter discount controls, validated against product cost and competitive conditions.
- **Review loss-exposed categories:** Men's Clothing, Health and Beauty, Video Games, Golf Shoes, and Pet Supplies should be reviewed for pricing, product cost, mix, and discount exposure even though they remain profitable in aggregate.

- Use differentiated market and regional strategies: Europe should be recognized for its large absolute contribution while USCA demonstrates stronger margin efficiency; Pacific Asia warrants closer review given its lower market-level margin and higher loss rate.
- Treat operational performance as a secondary lever: shipping and delivery performance should be monitored for customer experience, but given the small profitability differences observed, it should not be treated as the primary profitability-improvement lever.
- Integrate financial and operational analytics: the strongest long-term approach combines customer, product, category, discount, and geographic indicators within a single decision-support environment, building on the Streamlit dashboard described in Section 7.

## 7. STREAMLIT DASHBOARD IMPLEMENTATION

The analytical findings were operationalized through an interactive Streamlit dashboard titled "APL Logistics — Profitability & Supply Chain Dashboard", which transforms the static exploratory analysis into a decision-support interface organized into five modules.

- Revenue and Profit Overview — Total Sales, Total Profit, and Profit Margin KPIs with revenue-versus-profit comparisons; since the dataset lacks a usable order-date field, the dashboard avoids a false chronological trend and instead compares profitability across discount bands and markets.
- Customer Value Dashboard — total customer count, average profit per customer, top and bottom 10 customers by profit, customer value-tier contribution, segment contribution, and identification of loss-making customers.
- Product and Category Performance — product-level margin analysis, top product profitability, category-level profitability and margin comparison, loss-rate analysis, and heatmap-style category performance.
- Discount Impact Analyzer — discount-band margin and loss-rate comparison, average profit by discount level, high-discount category analysis, and exploratory what-if discount scenarios (an exploratory tool, not a predictive guarantee).
- Market and Regional Analysis — market-level sales, profit, and margin; regional profitability; country-level analysis where volume is sufficient; and a Revenue-vs-Profit-by-Market visualization.

Interactive controls allow filtering by Customer Segment, Category, Product, Market, Order Region, and Discount Rate, so stakeholders can investigate profitability under different conditions. The dashboard therefore functions as a practical extension of the research analysis, supporting the continuous-monitoring recommendation in Section 6.

## 8. CONCLUSION AND FUTURE WORK

### 8.1 Conclusion

This study demonstrates the importance of shifting supply chain analytics from a purely revenue-oriented perspective toward profitability intelligence. Analysis of 180,519 order items across 20,652 customers, 118 products, and 50 categories establishes that sales volume alone does not sufficiently represent business value: the dataset records 36.78 million in sales and 3.97 million in profit, an overall margin of 10.78%, yet substantial variation exists beneath this aggregate figure.

Customer analysis identifies 4,069 loss-making customers ( $\approx 19.70\%$  of the base), while product analysis reveals a far stronger concentration pattern in which the top 10 products contribute approximately 89.70% of total profit — indicating that customer and product

profitability require distinct management approaches. Discount analysis shows margin declining from 13.09% to 9.39% across discount bands, while loss rates fluctuate rather than rising monotonically. Market analysis shows commercial scale and profitability efficiency to be separate dimensions: Europe leads in absolute sales and profit, USCA records the highest market-level margin, and Pacific Asia records the lowest margin and highest loss rate. Operational analysis shows comparatively small profitability differences across shipping-performance groups, suggesting customer, product, and discount factors are stronger review priorities than shipping delay.

The Streamlit dashboard operationalizes these findings through interactive views of revenue, profit, customer, product, discount, and market performance. The central conclusion is that supply chain performance should be evaluated not by how much is sold, but by where revenue is generated, how much profit it produces, which customers and products create meaningful profit contribution, and where margins are being compressed. All relationships identified represent associations observed in historical data and should not be read as causal without further pricing, cost, and experimental evidence.

## **8.2 Future Work**

The following extensions could build on the current analysis as additional data becomes available:

- Predictive profitability and churn-risk modeling using customer, product, discount, and market variables to flag customers moving toward higher or lower profitability tiers.
- Discount optimization balancing sales-volume objectives with margin requirements at category or segment level.
- Cost-inclusive and time-series profitability modeling incorporating shipping, handling, freight, and product-cost data, once a reliable order-date field is available.
- An integrated supply chain intelligence platform building on the current dashboard as its foundation.

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